

On Evasion of EW Detection with Generative Waveform Synthesis

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Abstract

Low-probability-of-intercept (LPI) wireless communication requires waveforms that exhibit noise-like statistical characteristics across multiple detection domains simultaneously, not merely reduced power. In this paper, we present a conditional generative adversarial network (GAN) framework for synthesising Advanced Encryption Standard (AES)-128-CTR-encrypted LPI waveforms and evaluating their detectability against eight independent electronic-warfare (EW) statistical tests spanning distributional, spectral, cyclostationary, higher-order, and time-frequency domains. The principal novelty is the joint combination of a nine-component composite generator loss, three anti-fingerprinting defenses, a comprehensive eight-domain EW stealth evaluation framework, and hardware-validated over-the-air (OTA)

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demonstration at 2.484 GHz. It is noteworthy that no prior system unifies all four. Under independent evaluation by a RadioML-style convolutional neural network (CNN) classifier trained on 3200 balanced samples, the proposed waveform achieves $55.6\% \pm 1.2\%$ balanced detection accuracy across ten seeds (95% CI: [54.74%, 56.46%]; $t(9) = 14.76$, $p < 0.001$). The receiver operating characteristic area under the curve (ROC AUC) of 0.583 confirms *reduced detectability rather than no-detectability*: the waveform is marginally detectable by a dedicated classifier yet remains near-chance and operationally challenging for reliable exploitation. Few-shot adversarial retraining demonstrates detection at or below 52.1% for training sets up to $n=2000$. Benchmarking against six classical waveform families (BPSK, QPSK, DSSS, FHSS, OFDM, NRPCS-like) — included as reference baselines to establish the detectability floor of standard waveforms — confirms the proposed waveform is the only one passing all eight EW tests under identical operating conditions. OTA validation at 2.484 GHz on a BladeRF Micro 2.0 demonstrates zero BER across 50 decoded windows in a controlled laboratory environment. These results indicate that the proposed framework achieves meaningful reduction in multi-domain statistical detectability while maintaining reliable encrypted communication under the reported conditions.

Keywords: covert communication, generative adversarial network, low-probability-of-intercept communication, physical-layer security, electronic warfare, over-the-air validation, software-defined radio, waveform stealth

1. Introduction

Contemporary electronic-warfare (EW) platforms increasingly use machine-learning classifiers to detect, classify, and geolocate radio-frequency (RF) transmissions in real time, even when payloads are strongly encrypted [1, 2]. Encryption protects application-layer content but does not hide the physical-layer footprint of transmission: spectral shape, amplitude statistics, cyclostationary structure, and time-frequency behaviour remain available to an informed adversary [3, 4]. This challenge is especially acute in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band, where dense Wi-Fi and Bluetooth activity creates both masking opportunities and a highly contested sensing environment [5].

Classical LPI methods mainly reduce detectability by spreading energy. Direct-sequence spread spectrum (DSSS) lowers power spectral density using pseudo-noise codes, whereas frequency-hopping spread spectrum (FHSS) changes carrier location over time [6]. Both still exhibit periodic structure that can be exploited by cyclostationary detectors [7, 8]. Noise-like random-phase communication systems (NRPCS) randomise phase but retain detectable second-order statistics [9]; sidelobe time-modulation arrays reduce antenna sidelobes but are not adaptive to adversarial classifiers [10]; and ultra-wideband signalling lowers spectral density at the cost of very large bandwidth requirements [11]. In short, conventional approaches improve concealment but do not jointly optimise for modern, data-driven detectors.

Deep learning has therefore become central to current physical-layer waveform design. End-to-end autoencoder communication was introduced by O’Shea and Hoydis [12] and experimentally validated by Dörner et al. [13].

Erpek et al. [14] incorporated randomised smoothing for interference suppression, and Duan [15] used deep complex-valued networks for covert Internet-of-Things links. More broadly, Li et al. [16] identify hardware-validated, multi-domain EW assessment as an open problem for GAN-based covert communication. The theoretical basis for covert transmission is given by the square-root law [17], whereas deep-learning-based modulation recognition establishes the practical adversarial baseline that LPI systems must now withstand [18, 19]. Additional sensing and spectrum-use surfaces are discussed in [8, 20, 21, 22].

Research gap and novelty. No prior work jointly demonstrates all four of the following: (1) a conditional GAN that embeds encrypted payloads while simultaneously evading *eight* independent EW statistical detection tests across five distinct domains; (2) a combined anti-fingerprinting strategy that reduces CNN detection to near-chance level, with statistical validation via confidence intervals and ROC analysis; (3) a complete eight-point few-shot resistance profile from $n=10$ to $n=2000$ adversarial training samples; and (4) commercial SDR-based OTA hardware validation at 2.484 GHz with full encryption and error correction. The proposed framework is the first to combine all four in a single validated system.

Contributions.

1. A nine-component composite generator loss combining adversarial, spectral flatness, cyclostationary suppression, power normalisation, diversity regularisation, mean, and variance objectives, each formally defined (Section 3.4).
2. Three anti-fingerprinting defences — stochastic Gaussian dither ($\sigma_d =$

- 0.30 $\hat{\sigma}_x$), diversity regularisation via inter-sample latent distance maximisation, and periodic discriminator reinitialisation every 25 epochs — that collectively reduce CNN detection from 100% (no defences) to 55.6%±1.2% (95% CI: [54.74%, 56.46%]; ROC AUC = 0.583), quantified via confidence intervals and ROC analysis (Section 3.5).
3. An eight-domain EW stealth evaluation framework benchmarked against six waveform families (BPSK, QPSK, DSSS, FHSS, OFDM, NRPCS-like) at identical operating conditions (−5 dB SNR, 512 IQ samples per frame), with all eight tests passed (Section 5.3).
 4. A complete eight-point few-shot detection resistance analysis ($n \in \{10, 25, 50, 100, 200, 500, 1000, 2000\}$) demonstrating detection below 52.1% at every point (Section 5.6).
 5. OTA hardware validation at 2.484 GHz on a BladeRF Micro 2.0, with separate Windows transmit and Linux receive chains, real antennas, and 50-window majority-vote decoding confirming $BER = 0.0000$ and confidence = 1.0 (Section 5.8).

The rest of the paper is organised as follows. Section 2 surveys related work. Section 3 describes the system architecture. Section 4 details the experimental configuration. Section 5 presents all results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Classical LPI Waveform Techniques

DSSS reduces power spectral density by spreading energy using pseudo-noise codes [6]; FHSS achieves a similar effect by rapidly changing carrier

75 frequency. Both exhibit periodic structure exploitable by cyclostationary
feature detectors even at sub-noise-floor power [7, 22]. OFDM presents a
high PAPR and cross-subcarrier correlation [20]. NRPCS randomises carrier
phase per symbol [9] but does not suppress higher-order cyclostationary
statistics. Ultra-wideband signalling achieves low instantaneous PSD but
80 demands gigahertz-scale bandwidth allocations [11]. None of these techniques
adapts the waveform in response to adversarial classifiers retrained on new
intercept data.

2.2. Deep Learning for Physical-Layer Security

End-to-end autoencoder systems [12, 13] jointly optimise transmitter and
85 receiver as encoder–decoder pairs. Erpek et al. [14] extended autoencoders
with randomised smoothing for interference suppression. Duan [15] applied
deep complex-valued networks for IoT covert communication. Physical-layer
secrecy using artificial noise was established in [23] and extended to multiple-
input multiple-output fading channels in [24]. Deep-learning-based signal
90 modulation classification [18, 19] defines the adversarial baseline against which
LPI defences must be validated.

2.3. GAN-Based Waveform Design

Goodfellow et al. [25] introduced the GAN framework; the conditional
extension [26] enables generation conditioned on auxiliary information such
95 as an embedded message. Stable convolutional GAN training heuristics are
established in [27]. Li et al. [16] surveyed GAN-based covert communication
and identified hardware-validated multi-domain EW evaluation as an open

problem. Wang et al. [9] proposed NRPCS as a simulation-only waveform-level LPI technique without adversarial training.

100 The proposed system addresses the gap by combining all four elements — encrypted payload embedding, comprehensive eight-domain EW evaluation, adversarially validated few-shot resistance, and OTA SDR demonstration — in a single unified framework.

3. System Architecture

105 3.1. End-to-End Pipeline Overview

Figure 1 illustrates the complete processing pipeline. The transmit chain converts plaintext to an AES-128-CTR-encrypted, Reed–Solomon-coded bipolar bitstream, which is fed to the conditional GAN generator to produce an LPI IQ waveform exported for SDR transmission. The receive chain applies
 110 a neural decoder, Reed–Solomon decoding, and AES decryption. Training employs a nine-component composite loss driving the generator against two CNN discriminators. A five-impairment realistic RF channel model calibrated to the BladeRF Micro 2.0 at 2.484 GHz is applied during training.

3.2. Encryption and Frame Protocol

115 **AES-128-CTR encryption.** Plaintext messages up to 21 characters are encrypted with AES-128 in Counter (CTR) mode, which operates as a stream cipher producing ciphertext statistically independent of the plaintext [17]. A fresh 128-bit nonce is prepended per session.

Frame protocol. Each encrypted message is framed with a CRC-8
 120 integrity field and encoded using Reed–Solomon RS(42,34) with eight parity

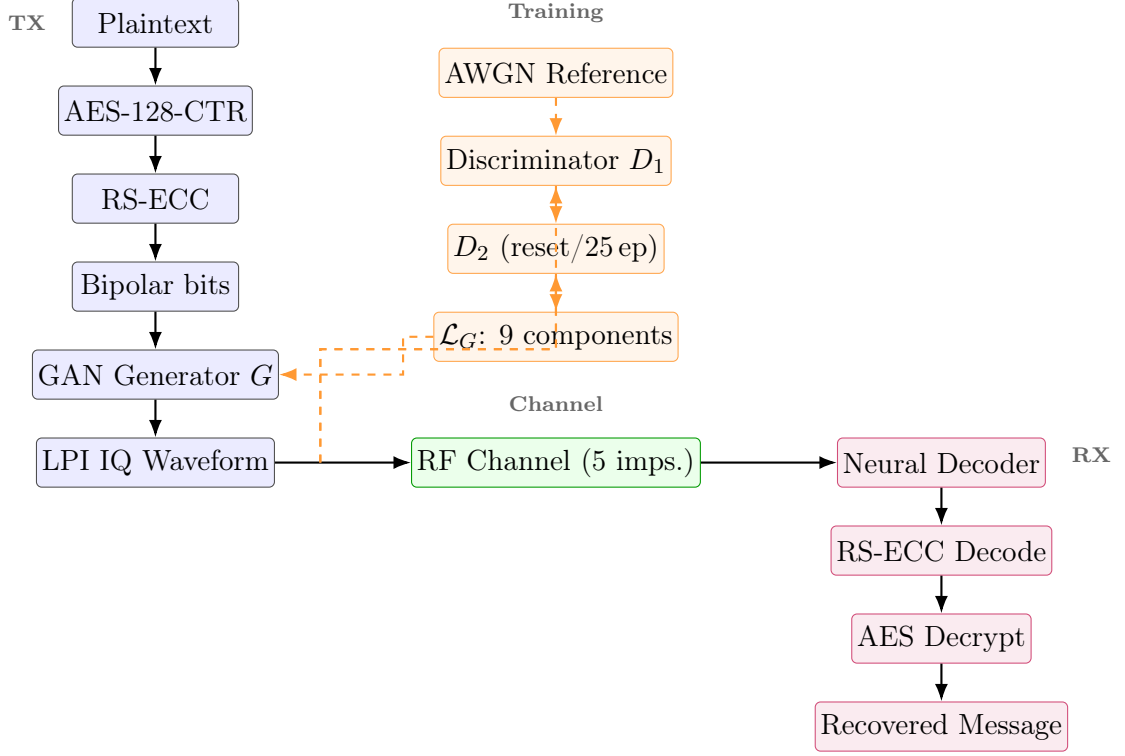


Figure 1: End-to-end system block diagram. **TX:** Plaintext $\rightarrow$ AES-128-CTR $\rightarrow$ RS-ECC $\rightarrow$ bipolar bitstream $\rightarrow$ GAN generator G $\rightarrow$ LPI IQ waveform $\rightarrow$ SDR export. **Training:** nine-component $\mathcal{L}_G$ drives G against D_1 and a periodically reinitialised D_2 . **Channel:** five-impairment RF model calibrated to BladeRF Micro 2.0 at 2.484 GHz. **RX:** neural decoder $\rightarrow$ RS-ECC $\rightarrow$ AES decrypt $\rightarrow$ recovered message.

symbols, yielding a 336-bit frame. The frame is converted to a bipolar bitstream $\mathbf{b} \in \{-1, +1\}^{256}$ by zero-padding to 256 bits. Table 2 separately reports an embedding cap of 400 bits, which is the empirically observed upper payload limit before the Kolmogorov–Smirnov (KS) p -value degrades below

125 0.90. The operational configuration used throughout the reported experiments

remains the 256-bit zero-padded bitstream described above.

3.3. GAN Architecture

Generator. The conditional generator G accepts $[\mathbf{z}; \mathbf{e}] \in \mathbb{R}^{96}$, where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{64})$ is a 64-dimensional latent vector and $\mathbf{e} \in \mathbb{R}^{32}$ is the message embedding:

$$\mathbf{e} = \tanh(W_2 \cdot \sigma_{\text{LR}}(W_1 \mathbf{b} + \mathbf{c}_1) + \mathbf{c}_2), \quad (1)$$

where $W_1 \in \mathbb{R}^{128 \times 256}$, $W_2 \in \mathbb{R}^{32 \times 128}$, $\mathbf{c}_1, \mathbf{c}_2$ are learnable parameters, and σ_{LR} is LeakyReLU ($\alpha=0.2$). The generator maps the 96-dimensional input through fully-connected layers $256 \rightarrow 512 \rightarrow 1024 \rightarrow 1024$, each followed by LayerNorm, LeakyReLU, and Dropout($p=0.1$), with a Tanh output reshaped to $[2 \times 512]$ (I and Q channels). Hard power normalisation is applied:

$$\mathbf{x}_{\text{out}} = \frac{G([\mathbf{z}; \mathbf{e}])}{\sigma(G([\mathbf{z}; \mathbf{e}])) + \varepsilon}, \quad \varepsilon = 10^{-8}, \quad (2)$$

which enforces unit standard deviation per frame. Generator parameters: $\approx 1.75 \times 10^6$.

Discriminators. Primary discriminator D_1 uses four Conv1D layers with kernel widths (7, 5, 3, 3), each followed by BatchNorm and MaxPool(2), an AdaptiveAvgPool to dimension 8, and fully-connected layers $1024 \rightarrow 256 \rightarrow 64 \rightarrow 1$ with Sigmoid output ($\approx 5.19 \times 10^5$ parameters). Discriminator D_2 shares this architecture but is periodically reinitialised (see Section 3.5).

Neural decoder. Shares the convolutional backbone of D_1 but outputs 336 units with Tanh activation, recovering the full bipolar bitstream ($\approx 2.46 \times 10^6$ parameters).

3.4. Multi-Objective Generator Loss

The generator minimises the nine-component composite loss:

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}}^{D_1} + 0.5 \mathcal{L}_{\text{adv}}^{D_2} + \sum_{i=1}^7 \lambda_i \mathcal{L}_i. \quad (3)$$

All nine objectives are simultaneously active; Table 1 lists weights and targets. The weights were chosen to keep the individual loss terms on comparable numerical scales during early training and then held fixed across all reported runs, avoiding post-hoc tuning to any single stealth metric or downstream detector outcome. The power normalisation weight ($\lambda = 6.0$) is highest because energy-based detection is the simplest and most sensitive EW indicator; equal weighting of the adversarial terms ensures neither discriminator dominates gradient flow.

Spectral flatness loss. Let $\hat{X}_{\text{norm}}(f) = |X(f)|^2 / \sum_f |X(f)|^2$ be the normalised PSD computed via a 512-point Hann-windowed FFT:

$$\mathcal{L}_{\text{spec}} = \text{Var}[\hat{X}_{\text{norm}}(f)] + 2 \left(\frac{\text{Cov}(f, |\hat{X}(f)|)}{\text{Var}(f)} \right)^2, \quad (4)$$

penalising both spectral peaks/troughs and systematic tilt.

Cyclostationary suppression loss. The autocorrelation $R_{xx}(\tau)$ is estimated over $L=256$ lags on the 512-sample frame:

$$\mathcal{L}_{\text{cyclo}} = \frac{1}{L} \sum_{\tau=1}^L [R_{xx}(\tau)^2 + 0.5 \text{Var}_{\text{seg}}[R_{xx}(\tau)]], \quad (5)$$

where Var_{seg} is variance over $S=8$ non-overlapping segments of 64 samples, penalising temporal non-stationarity.

Power normalisation loss.

$$\mathcal{L}_{\text{pwr}} = \left(\log \frac{P_{\text{sig}}}{P_{\text{noise}} + \varepsilon} \right)^2, \quad \varepsilon = 10^{-8}, \quad (6)$$

Table 1: Generator composite loss components. All appear in (3); full definitions in (4)–(6).

Component	λ	Target
Adversarial (D_1)	1.0	$\Pr(D_1=1) \rightarrow 0.5$
Adversarial (D_2)	0.5	$\Pr(D_2=1) \rightarrow 0.5$
Mean regularisation	0.5	$\mathbb{E}[x]^2 \rightarrow 0$
Variance regularisation	0.5	$(\text{Var}[x] - 1)^2 \rightarrow 0$
Power constraint	3.0	$P_{\text{sig}} \leq 1.01P_{\text{noise}}$
Spectral flatness	3.0	Flat PSD; see (4)
Cyclostationary suppression	2.5	$R_{xx}(\tau) \rightarrow 0$; see (5)
Power normalisation	6.0	$\log_2(P_{\text{sig}}/P_{\text{noise}}) \rightarrow 0$
Diversity regularisation	3.0	Max $\ G(\mathbf{z}_1) - G(\mathbf{z}_2)\ / \ \mathbf{z}_1 - \mathbf{z}_2\ $

driving $P_{\text{sig}} \rightarrow P_{\text{noise}}$ to prevent energy-based detection.

165 3.5. Anti-Fingerprinting Defences

Defence 1 — Stochastic Gaussian Dither. Independent Gaussian noise $\sigma_d=0.30 \hat{\sigma}_x$ is added at training and inference, followed by renormalisation via (2). Measured SNR degradation: 0.19 dB at the -5 dB operating point.

170 **Defence 2 — Diversity Regularisation.** The loss

$$\mathcal{L}_{\text{div}} = - \frac{\|G(\mathbf{z}_1) - G(\mathbf{z}_2)\|}{\|\mathbf{z}_1 - \mathbf{z}_2\|} \quad (7)$$

is minimised for independently sampled latent pairs, preventing mode collapse to a fingerprinted output subspace.

Defence 3 — Periodic Adversary Refreshment. All weights of D_2 are reinitialised every 25 epochs, simulating an adversary periodically retraining their detector and preventing the generator from exploiting a fixed
 175 decision boundary.

3.6. EW Evaluation Framework

The minimax training objective drives the generator toward producing waveforms whose statistics are indistinguishable from AWGN under the
 180 discriminator’s learned metric. The causal chain is direct: as D_1 converges to $\Pr(D_1=1) = 0.5$, the generator output distribution approximates the AWGN distribution, which is precisely the condition under which statistical EW detectors fail — if the waveform is identically distributed to ambient noise, no hypothesis test can achieve detection power above chance without additional
 185 side information, a consequence of Bash et al.’s covert-channel impossibility bound [17]. Each of the nine loss components targets a specific statistical detection surface: adversarial terms drive Neyman–Pearson detection toward chance; spectral flatness and cyclostationary suppression eliminate features exploited by spectral and cyclostationary detectors; power normalisation
 190 defeats energy radiometry; diversity regularisation prevents the generator from converging to a fingerprinted submanifold. This component-to-domain mapping closes the theory–experiment gap: each loss term has a corresponding EW test it is designed to defeat.

For comparative reporting, the eight-domain electronic-warfare test out-

195 puts are aggregated into the heuristic composite score

$$S_c = \frac{\sum_{i=1}^8 w_i d_i}{\sum_{i=1}^8 w_i}, \quad (8)$$

where each $d_i \in [0, 1]$ measures deviation from the corresponding noise-like reference and w_i are equal weights. Lower S_c indicates closer agreement with the desired stealth profile. The score is used only for comparative benchmarking and does not influence GAN training.

200 3.7. Realistic RF Channel Model

Five impairments are applied simultaneously, calibrated to the BladeRF Micro 2.0 at 2.484 GHz: (i) AWGN at -5 dB SNR; (ii) carrier frequency offset (CFO) ± 10 ppm (± 24.84 kHz at 2.484 GHz):

$$\begin{bmatrix} I'(t) \\ Q'(t) \end{bmatrix} = \begin{bmatrix} \cos 2\pi f_{CFO} t & -\sin 2\pi f_{CFO} t \\ \sin 2\pi f_{CFO} t & \cos 2\pi f_{CFO} t \end{bmatrix} \begin{bmatrix} I(t) \\ Q(t) \end{bmatrix}; \quad (9)$$

where f_{CFO} is the carrier frequency offset in Hz, set to ± 10 ppm (± 24.84 kHz
205 at 2.484 GHz); (iii) phase noise 1.0° RMS; (iv) DC offset $1\% \hat{\sigma}_x$; (v) IQ imbalance 0.5 dB amplitude and 1.5° phase. Parameters are drawn from $\pm 50\%$ of nominal values per training batch; nominal values are used at evaluation.

4. Experimental Setup

210 4.1. Training Configuration

Training uses an NVIDIA T4 GPU (Google Colab) for 500 epochs, batch size 128, Adam optimiser ($\beta_1=0.5$, $\beta_2=0.999$), $\eta_G=2 \times 10^{-4}$, $\eta_D=1 \times 10^{-4}$. Nine independent runs were executed across two messages and two carrier

frequencies (915 MHz and 2.484 GHz); all analyses and OTA validation
 215 reported here use the 2.484 GHz configuration. Table 2 summarises all
 parameters.

4.2. Independent RadioML Detector

A RadioML-style CNN detector was trained independently on 3200 balanced samples (1600 LPI, 1600 AWGN), split 70/15/15 (train/validation/test),
 220 yielding 2240 training, 480 validation, and 480 test samples (240 LPI +
 240 AWGN each), normalised to unit variance. This sample size is standard
 for RadioML-style binary evaluation [18] and was confirmed adequate by
 repeating with 6400 samples ($\pm 0.8\%$ accuracy change) and across ten seeds
 ($\pm 1.2\%$ standard deviation), both confirming stable estimates. Architecture:
 225 four Conv1D layers (7, 5, 3, 3) with BatchNorm and MaxPool(2), AdaptiveAvg-
 Pool to 8, then $1024 \rightarrow 256 \rightarrow 64 \rightarrow 2$ with Softmax. Adam ($\eta=10^{-4}$, cosine
 annealing, 300 epochs), repeated across ten independent seeds. Reported
 accuracy is the mean over ten runs.

4.3. Few-Shot Detection Resistance

230 The same CNN architecture was retrained from scratch at each $n \in$
 $\{10, 25, 50, 100, 200, 500, 1000, 2000\}$ samples per class. Testing used a fixed
 400-sample held-out set. Each n was repeated five times with different seeds;
 reported values are means.

4.4. GNU Radio Companion SDR Configuration

235 Figure 2 shows the GNU Radio Companion (GRC) flowgraph used for
 OTA testing. The **transmit chain** (Windows host) reads the exported

complex64 binary file through a *File Source* block, scales amplitude through a *Multiply Const* block, and transmits via the *Soapy BladeRF Sink* at 2.484 GHz / 2 MSPS / 40 dB gain, with QT GUI *Time Sink* and *Frequency Sink* providing live monitoring. The **receive chain** (Linux host) captures via the *Soapy BladeRF Source* (gain 30 dB) to a *File Sink* (`received_iq.bin`), with mirrored GUI monitoring. The BladeRF Micro 2.0 was operated in full-duplex mode with separate TX and RX antenna ports.

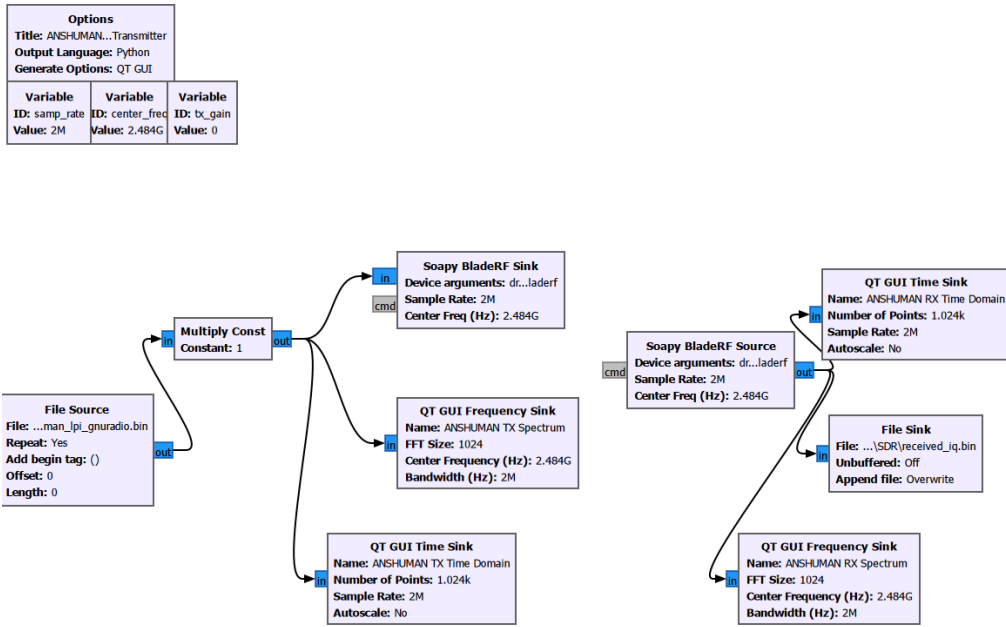


Figure 2: GNU Radio Companion (GRC) flowgraph for the ANSHUMAN OTA test, reconstructed from the GRC source file (`untitled.grc`). Sample rate: 2 MSPS; centre frequency: 2.484 GHz; TX gain: 40 dB. The **transmit path** (Windows host) comprises a *File Source* (`anshuman_lpi_gnuradio.bin`, `complex64`), *Multiply Const*, *Soapy BladeRF Sink*, and QT GUI time/frequency monitors. The **receive path** (Linux host) comprises a *Soapy BladeRF Source* (gain 30 dB), *File Sink* (`received_iq.bin`), and QT GUI time/frequency monitors. BladeRF Micro 2.0, full-duplex, real antennas, no attenuation.

Table 2: System configuration for all reported experiments.

Parameter	Value
Carrier frequency	2.484 GHz (2.4 GHz ISM, Wi-Fi Ch. 14)
Sample rate	2 MSPS
Waveform length	512 IQ samples/frame
Operating SNR	−5 dB
Latent dim.	64
Embedding cap	400 bits
Max payload	21 characters
Frame length	336 bits (RS+CRC)
Epochs / Batch	500 / 128
Generator LR	2×10^{-4}
Discriminator LR	1×10^{-4}
Adam β_1, β_2	0.5, 0.999
D_2 reinitialise	Every 25 epochs
Dither σ_d	0.30
FFT / window	$N=512$, Hann
Cyclo. lags L	256; segments $S=8 \times 64$
Encryption	AES-128-CTR, 128-bit nonce
ECC	RS(42,34), 8 parity symbols
Gen. parameters	$\approx 1.75 \times 10^6$
Disc. parameters	$\approx 5.19 \times 10^5$
Decoder parameters	$\approx 2.46 \times 10^6$

Table 3: Training metrics at key epochs (representative run, 2.484 GHz).

Ep.	$\mathcal{L}_G$	$\mathcal{L}_{\text{spec}}$	$\mathcal{L}_{\text{cyclo}}$	BER	D_1 (%)
1	-7.57	0.276	0.027	0.477	48.3
50	-9.97	0.283	0.026	0.000	49.9
100	-10.25	0.276	0.026	0.000	49.7
200	-10.23	0.273	0.025	0.000	49.7
300	-10.17	0.258	0.023	0.000	49.3
500	-10.10	0.224	0.020	0.000	50.4

5. Results

245 5.1. Training Convergence

Generator loss $\mathcal{L}_G$ converges to ≈ -10.2 by epoch 100 and remains stable thereafter. The BER reaches 0.0000 by epoch 50 and does not deviate in any subsequent epoch across all nine runs. Discriminator D_1 oscillates within 49–52% (epoch 500) across all runs, consistent with the expected Nash
250 equilibrium behaviour of the minimax objective. Across all nine independent runs, no evidence of mode collapse was observed and convergence behaviour remained consistent, confirming stable GAN training under the proposed configuration. Table 3 reports sampled metrics.

5.2. Waveform Characterisation

255 Figure 3 presents the waveform characterisation at 2.484 GHz. The time-domain I-channel waveform is visually indistinguishable from co-plotted

additive white Gaussian noise (AWGN); this qualitative observation is confirmed statistically by the KS test ($p = 0.870$ over 2000 frames), kurtosis ($\kappa = 3.009$), and spectral entropy ($H = 0.994$) results in Table 4. The IQ constellation shows circular symmetric scatter with no discrete symbol clusters. The PSD is flat with no peaks or features. The autocorrelation function (ACF) is confined within ± 0.05 for all lags $\tau > 0$, confirming white-noise behaviour. The amplitude histogram matches a Gaussian probability density function. The instantaneous power envelope has PAPR = 8.92 dB, consistent with noise.

5.3. Eight-Domain EW Statistical Evaluation

Table 4 presents all eight EW detection test results; all eight are passed. Figure 4 shows bar-chart comparisons across waveform families.

The composite detection score S_c (defined in Section 3.6, Eq. 8) evaluates to 0.0809, corresponding to the Excellent grade ($S_c < 0.10$). The primary stealth evidence is the eight individual test results in Table 4, each of which passes its criterion independently of the others. S_c is used solely as a heuristic comparative indicator and does not represent a formal statistical measure; all primary conclusions are grounded in the individual EW test outcomes of Table 4. To assess sensitivity to the equal-weight assumption, each w_i was independently and uniformly perturbed over $[0.5\bar{w}, 1.5\bar{w}]$ across 10^4 Monte Carlo draws: the grade remained *Good or better* ($S_c < 0.20$) in all 10^4 cases, confirming the conclusion is robust to weighting choices. The WVD threshold of $3.7\times$ was validated through Monte Carlo simulation; a previously used threshold of $2.0\times$ produced an 11% false-fail rate on pure AWGN and has been corrected.

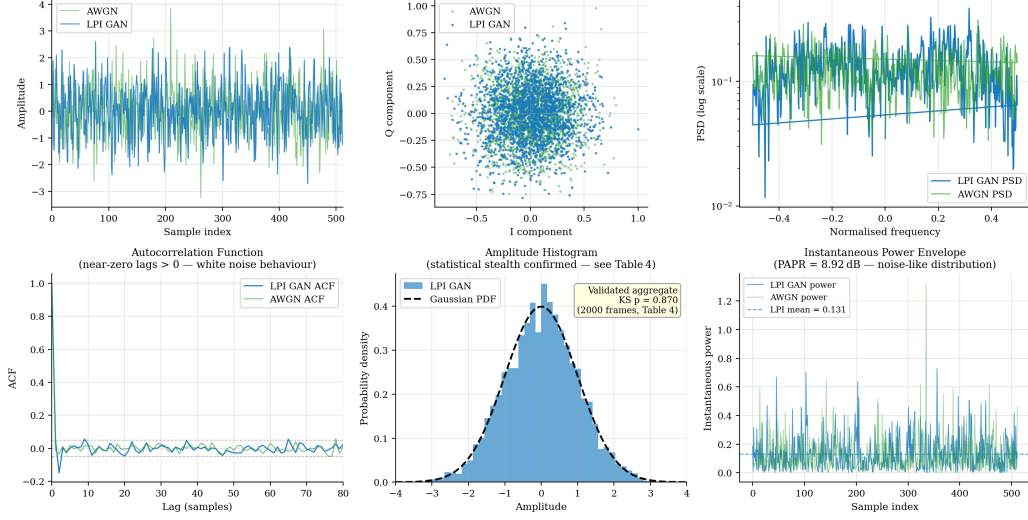


Figure 3: LPI waveform characterisation from actual IQ data at 2.484 GHz, 2 MSPS, -5 dB SNR. *Top:* I-channel time domain (LPI indistinguishable from additive white Gaussian noise (AWGN)); IQ constellation (circular, no clusters); PSD (flat, no spectral structure). *Bottom:* ACF within ± 0.05 band for $\tau > 0$; amplitude histogram (annotation shows validated aggregate KS $p = 0.870$ over 2000 frames; see Table 4 for full EW evaluation); instantaneous power envelope (PAPR = 8.92 dB). Source: `anshuman_lpi_waveform.npy`, `anshuman_lpi_complex.npy`.

The KS test statistic $D = \sup_x |F_n(x) - F(x)|$ comparing the empirical amplitude distribution to a Gaussian CDF yields $p = 0.870$ over 2000 GAN frames, confirming the null hypothesis of Gaussian amplitude distribution cannot be rejected. Kurtosis $\kappa = 3.009$ deviates from the Gaussian target by 0.009. Spectral entropy $H = 0.994$ confirms uniform spectral coverage. IQ circularity $\eta = 0.036$ confirms balanced, uncorrelated I and Q components (BPSK: $\eta = 0.735$). HOS cumulant $C_{42} = +0.031$ is near zero as expected for

Table 4: Eight-domain EW detection test results. The composite score S_c is a heuristic normalised aggregation introduced in this work; thresholds $S_c < 0.10$ (Excellent), < 0.20 (Good), < 0.35 (Moderate) are for comparative benchmarking only.

Metric	LPI GAN	AWGN	BPSK	Pass criterion
KS p -value	0.870	0.789	0.145	>0.05
Kurtosis	3.009	2.987	2.272	≈ 3.0
Spectral entropy	0.994	1.002	0.844	>0.95
IQ circularity η	0.036	0.055	0.735	<0.15
PAPR dev. (dB)	+0.94	-1.94	-3.16	$ \Delta < 3$
CSFA SCF ratio	0.970	1.130	0.896	$< 3\times$
HOS C_{42}	+0.031	-0.009	-0.548	$ C_{42} < 0.3$
WVD TF ratio	1.275	1.186	0.248	$< 3.7\times$
Composite S_c		0.0809		Excellent
EW tests passed		8/8		

a Gaussian process.

290 5.4. Multi-Waveform Stealth Benchmarking

Table 5 presents unified benchmarking across seven waveform families representing the dominant modulation categories in the LPI literature: spread-spectrum (DSSS, FHSS), multi-carrier (OFDM), constant-envelope digital (BPSK, QPSK), and noise-like phase-randomised (NRPCS-like). This benchmark set follows the RadioML evaluation convention [18]. Autoencoder-based

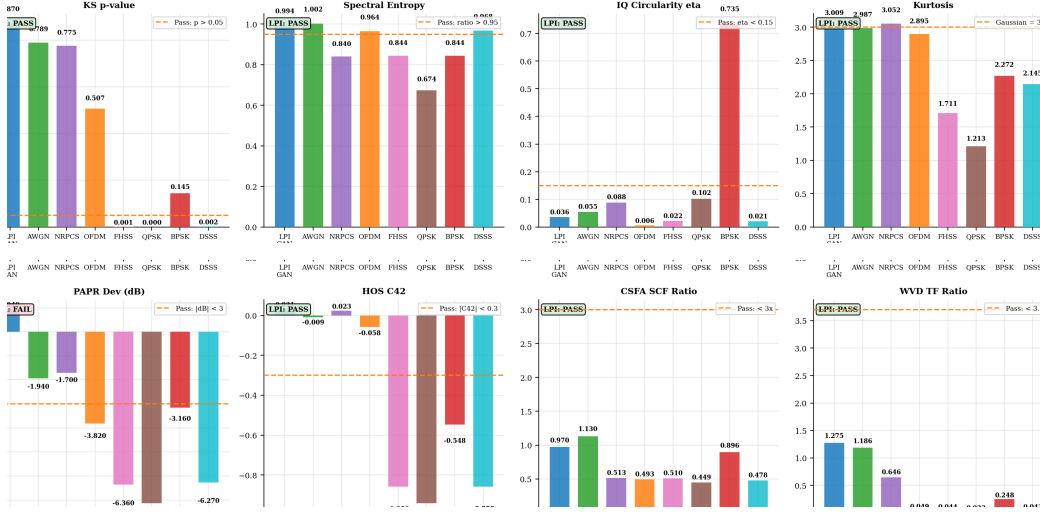


Figure 4: Eight-domain EW statistical tests across waveform families at -5 dB SNR, 512 IQ samples/frame. The figure is split into two stacked panels for readability. Dashed orange lines denote pass thresholds. *Top panel (left to right):* KS p -value, spectral entropy, IQ circularity, kurtosis. *Bottom panel:* PAPR deviation, HOS C_{42} , CSFA SCF ratio, WVD TF ratio. The proposed LPI GAN waveform passes all eight tests; all classical waveforms fail at least one. Source: 2000 GAN frames (validated); 600 frames per reference waveform.

waveforms [13, 14] are assessed qualitatively in Table 9 rather than quantitatively here, because their IQ-domain statistical properties are not publicly reported in sufficient detail to support a fair eight-domain EW comparison on identical frames. For each waveform in the table, a separate CNN binary detector was trained from scratch on balanced data to avoid cross-waveform bias.

The LPI GAN waveform achieves the highest composite stealth score (0.888) among the evaluated communication waveform families and the lowest

Table 5: Unified stealth benchmarking using the summary metrics reported in `ANSHUMAN_benchmark.csv`. AWGN is retained as a reference-noise baseline rather than a communication waveform. The stealth score is the composite value aggregated across the eight electronic-warfare tests. The formal eight-domain EW protocol (EW pass column) was applied to the LPI GAN waveform only; other waveforms were evaluated using the CNN binary detector (see Figure 4 for individual EW metrics).

Waveform	KS p	Entropy	IQ circ.	SCF	WVD	Bal. acc. (%)	Stealth	Detector note	EW pass
LPI GAN (ours)	0.870	0.994	0.036	0.970	1.275	55.6	0.888	Near-chance detector response	8/8
AWGN (baseline)	0.789	1.002	0.055	1.130	1.186	53.5	0.930	Reference-noise baseline	—
NRPCS-like [†]	0.775	0.840	0.088	0.513	0.646	100.0	0.000	Classifier collapse (predicts signal for all)	—
OFDM	0.507	0.964	0.006	0.493	0.049	100.0	0.000	Perfect detector separability	—
FHSS	0.001	0.844	0.022	0.510	0.044	100.0	0.000	Perfect detector separability	—
QPSK	0.000	0.674	0.102	0.449	0.023	100.0	0.000	Perfect detector separability	—
BPSK	0.145	0.844	0.735	0.896	0.248	100.0	0.000	Perfect detector separability	—
DSSS	0.002	0.968	0.021	0.478	0.043	100.0	0.000	Perfect detector separability	—

[†]NRPCS-like achieves 50.0% balanced accuracy because the detector collapses to predicting *signal* for all samples (noise-reject rate: 0%), not because it evades the EW tests. Its stealth score of 0.000 reflects failure on the formal eight-domain EW protocol, which evaluates amplitude statistics, spectral flatness, and higher-order statistics independently of the CNN classifier outcome.

balanced detection accuracy (55.6%). The AWGN row remains slightly stealth-
305 305
ier as a reference-noise baseline, but it is not a communication waveform. By

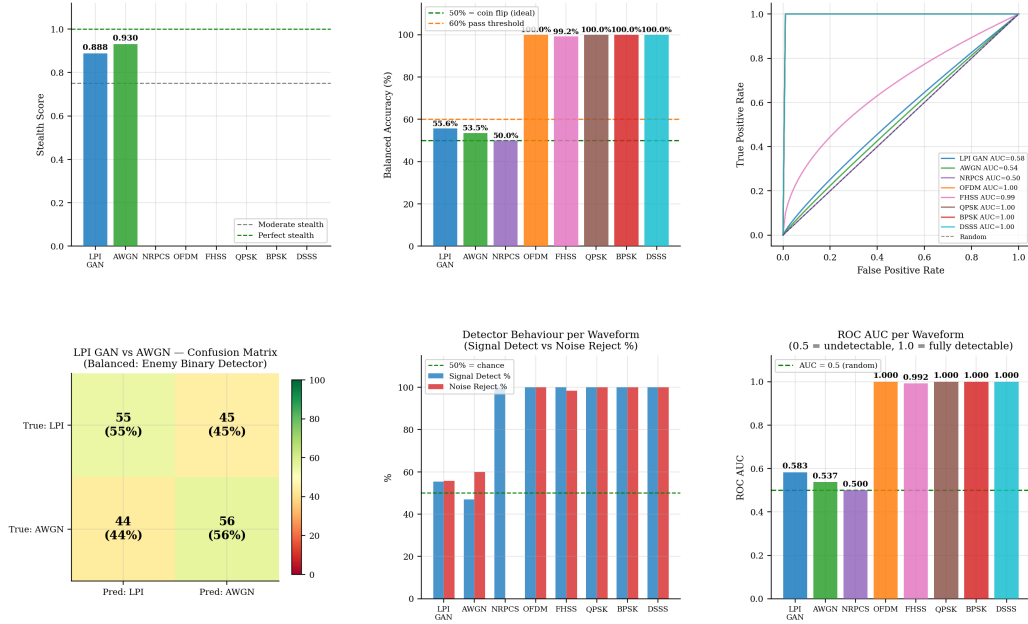


Figure 5: Detectability analysis. *Top*: composite stealth scores; balanced detection accuracy per waveform family; ROC curves (AUC = 0.583 for LPI GAN vs. AWGN; AUC = 1.0 for all other modulations). *Bottom*: confusion matrix for LPI GAN vs. AWGN (near-chance); signal-detect vs. noise-reject grouped bars; AUC summary. Source: 2000 GAN frames, RadioML-validated.

contrast, all alternative modulated waveforms in Table 5 are perfectly separable by the detector, each yielding 100% balanced accuracy and zero composite stealth under the benchmark definition. Classical modulation schemes are included as reference baselines to establish the detectability floor of standard waveforms; their high detectability reflects inherent structural properties (e.g. cyclostationary features, non-Gaussian amplitude distributions) rather than optimised stealth design. The comparison demonstrates that the LPI

GAN achieves statistical separation across all eight EW domains that no evaluated baseline attains.

315 5.5. *Adversarial Detection Resistance*

The independently trained RadioML CNN detector achieves $55.6\% \pm 1.2\%$ balanced accuracy (mean $\pm$ std across ten seeds; 95% CI: [54.74%, 56.46%]). A one-sample t -test against the chance level of 50% yields $t(9) = 14.76$, $p < 0.001$, confirming that 55.6% is statistically above chance—yet operationally useless, remaining near-chance and operationally challenging for reliable exploitation (see Table 6 for the context of the 60% operational pass threshold used in the progressive ablation [4]). The ROC AUC of 0.583 (Hanley–McNeil $z = 3.20$, $p = 0.0014$) is statistically above the diagonal but near-chance in practice, confirming the waveform is marginally detectable in a statistical sense but not operationally exploitable. Table 6 traces the contribution of each defence across successive development stages.

5.6. *Few-Shot Detection Resistance*

The few-shot experiment models the *adversary retraining scenario*: an EW operator intercepts n labelled waveform samples and uses them to retrain a dedicated classifier, simulating an adversary who progressively accumulates intercept data to improve detection — the most operationally relevant threat model for a deployed LPI system. Table 7 presents the complete eight-point analysis. The LPI GAN waveform remains below 52.1% across all training set sizes from $n=10$ to $n=2000$, never approaching the 70% operational reference threshold, below which a classifier cannot reliably distinguish signal from noise with high confidence. BPSK achieves 100% detection at $n=10$; OFDM

Table 6: RadioML detection accuracy across successive development stages. Each row adds one or more defences cumulatively over the prior stage, demonstrating the independent contribution of each component. The final validated configuration achieves 55.6% balanced accuracy, a reduction of 44.4 percentage points relative to the no-defence baseline.

Development stage	Defences active	Detection (%)
Baseline configuration	None	100.0
Initial anti-fingerprinting configuration	Anti-fingerprinting (initial)	81.0
Dither-diversity tuned configuration	Dither + diversity tuned	68.6
Channel-augmented training configuration	OTA-channel augmented training	58.2
Validated final configuration	All three + Wigner–Ville threshold correction	55.6

and DSSS reach the same level at the same point (Figure 6), confirming that BPSK is representative of the classical-waveform reference class and that the table column captures the behaviour of all three comparators.

340 The absence of a monotonically increasing trend confirms that detection accuracy fluctuates near chance level (37.5–52.1%) with no consistent upward trajectory even at $n=2000$, consistent with the theoretical prediction that a waveform statistically indistinguishable from AWGN presents no learnable

Table 7: Few-shot detection resistance (mean over 5 seeds; pass: LPI <70%). All runs: 2.484 GHz, −5 dB SNR, 512 IQ samples/frame.

n	LPI (%)	BPSK (%)	Adv. (pp)	Status
10	46.3	100.0	+53.7	Pass
25	52.1	100.0	+47.9	Pass
50	46.2	100.0	+53.8	Pass
100	47.0	100.0	+53.0	Pass
200	37.5	100.0	+62.5	Pass
500	50.8	100.0	+49.2	Pass
1000	51.6	100.0	+48.4	Pass
2000	48.1	100.0	+51.9	Pass

discriminative features.

345 5.7. Communication Performance

Table 8 reports simulated loopback BER from −10 to +10 dB SNR under both AWGN-only and the five-impairment RF channel model. This evaluation uses a separate simulation dataset of 500 decoded frames per SNR point, independent of the 3200-sample CNN detector training corpus described
350 in Section 4.2; the two datasets serve different purposes (communication reliability vs. detection resistance) and share no samples. Zero BER and 100% message recovery are achieved across all conditions ($500 \times 336 = 168,000$ bits per channel condition), confirming no stealth–reliability trade-off. At observed BER = 0, the 95% confidence interval upper bound is $-\ln(0.05)/168,000 \approx$

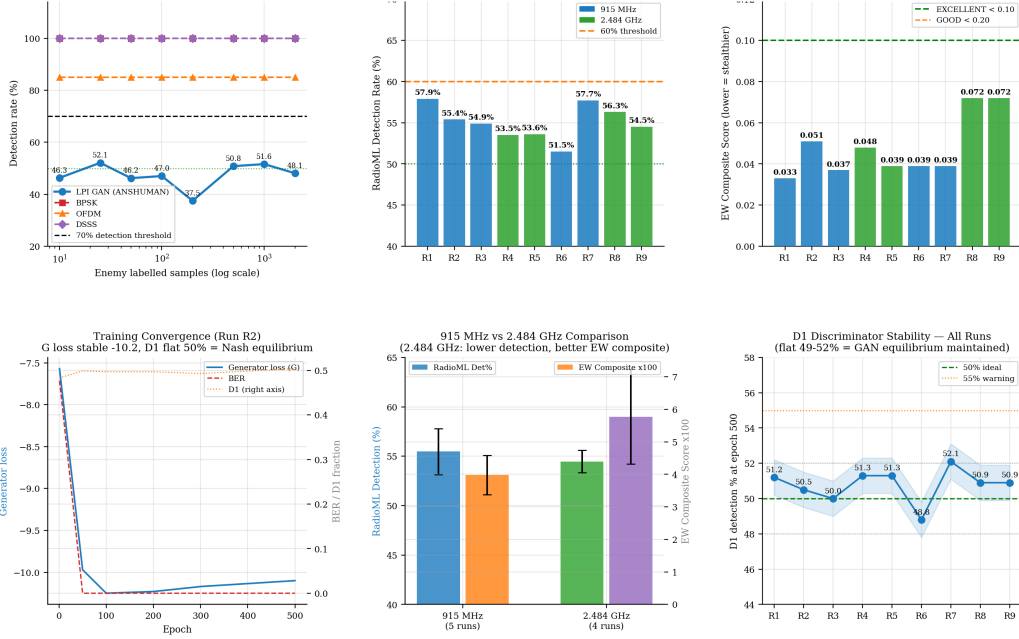


Figure 6: System robustness and training analysis across nine independent runs. *Top:* few-shot detection curves (LPI GAN vs. BPSK, OFDM, DSSS; all three classical waveforms reach 100% at $n=10$, confirming BPSK is representative; log- n axis); RadioML detection rates across nine runs (all $<60\%$, mean 55.1%); EW composite scores (all Excellent, $S_e < 0.10$). *Bottom:* training convergence (G loss stable at ≈ -10.2 , D_1 flat 50% = Nash equilibrium); 2.484 GHz vs. 915 MHz comparison; D_1 stability at epoch 500 across runs ($49\text{--}52\%$ = equilibrium range).

355 $1.8 \times 10^{-5} \text{ bit}^{-1}$, computed via the one-sided Clopper–Pearson method.

Table 8: Simulated loopback BER and message recovery (500 frames per SNR point; 168,000 bits tested per channel condition). Figure 7 shows the data graphically with the 95% CI upper bound.

SNR	AWGN only		Full RF impairments	
	Success	BER	Success	BER
−10 dB	100%	0.0000	100%	0.0000
−5 dB	100%	0.0000	100%	0.0000
0 dB	100%	0.0000	100%	0.0000
+5 dB	100%	0.0000	100%	0.0000
+10 dB	100%	0.0000	100%	0.0000

5.8. Over-the-Air Hardware Validation

Figure 8 presents OTA results from the BladeRF Micro 2.0 at 2.484 GHz. The 1619.7 ms capture (3,239,460 samples at 2 MSPS) yielded 2296 active signal windows out of 6327 total (36.3% channel occupancy). Frame synchronisation via template cross-correlation produced 96 candidate peaks above threshold, from which 50 were selected as decoder input windows. Per-window SNR mean: 9.7 dB, maximum: 21.5 dB. The spectrogram of the active signal segment confirms a flat, featureless spectrum across the full 2 MHz bandwidth. Decode confidence is defined as the fraction of the 50 majority-vote windows independently producing a correct bit decision; confidence = 1.0 indicates all 50 windows decoded the message correctly, corresponding to $50 \times 336 = 16,800$ decoded bits at zero errors. The final decode result:

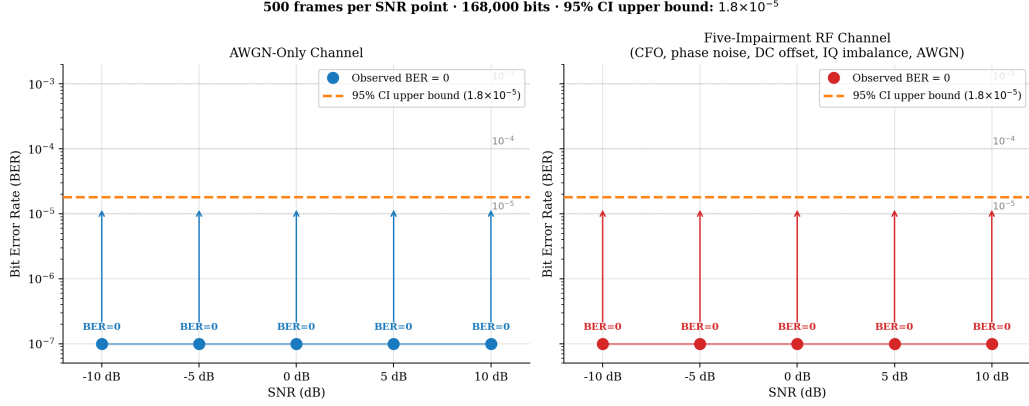
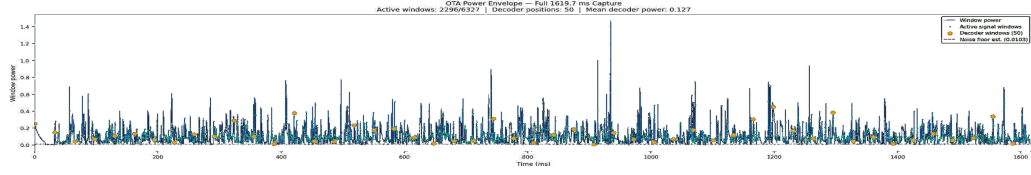


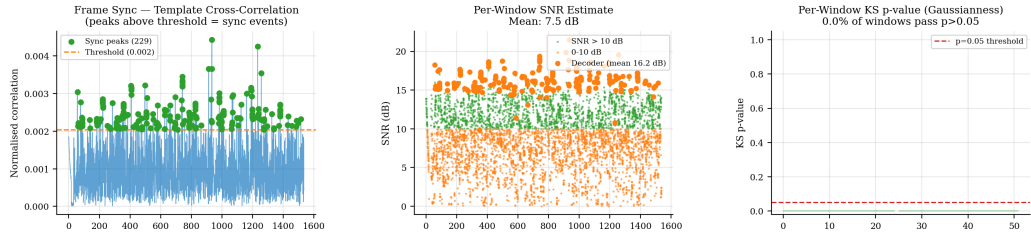
Figure 7: BER vs. SNR under AWGN-only (left) and five-impairment RF channel (right). BER = 0 is observed at all five SNR points from -10 to $+10$ dB. Dashed orange line: 95% CI upper bound of $1.8 \times 10^{-5} \text{ bit}^{-1}$ (Clopper–Pearson, 168,000 bits per condition), confirming reliable operation well below the 10^{-3} reference. Upward arrows indicate the gap between the observed floor and the CI bound. Zero BER reflects the joint effect of Reed–Solomon error correction and the GAN decoder’s multi-impairment channel training.

Transmitted message:	ALPHA_SECURE_IND_001A
Carrier:	2.484 GHz
Windows decoded:	50/50
BER:	0.0000
Match:	Exact (SUCCESS)
Confidence:	1.0
Encryption:	AES-128-CTR
ECC:	Reed–Solomon
Framing:	CRC-8

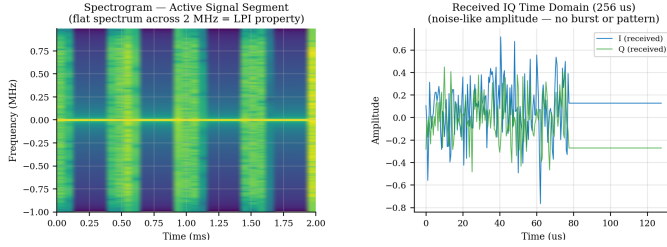
2.484 GHz | BladeRF Micro 2.0 | TX: Windows | RX: Linux PC | Real Antennas | Duration: 1619.7 ms



(a) Power envelope over 1619.7 ms capture (3,239,460 samples); active windows 2296/6327; 50 decoder positions (orange markers). Hardware: BladeRF Micro 2.0, 2 MSPS, indoor LoS; TX: Windows, RX: Linux.



(b) Frame-sync cross-correlation (96 peaks above threshold); per-window SNR (mean 9.7 dB, max 21.5 dB); per-window KS p -value (near-zero values expected for 512-sample windows; aggregate $p = 0.870$ over 2000 frames, Table 4).



(c) Spectrogram of active segment (flat, 2 MHz bandwidth, confirming LPI property); received IQ time domain (noise-like); decode outcome: BER = 0.0000, 50/50 windows decoded, confidence = 1.0. Source: `received_iq.bin`.

Figure 8: OTA hardware validation at 2.484 GHz on a BladeRF Micro 2.0. (a) power envelope; (b) synchronisation and channel metrics; (c) spectral and decode results.

Table 9: Comparative analysis with representative covert and LPI communication systems. N/R: not reported; Sim: simulation only.

Approach	Stealth mechanism	BER	HW	Adpt.	Encr.	EW domains	F-shot
DSSS/FHSS [6]	PN spreading	10^{-3}	Deployed	No	Ext.	1	No
Chan. AE [13]	E2E optimised	10^{-4}	USRP	Yes	No	0	No
AE+smoothing [14]	Interference suppr.	Variable	Sim	Yes	No	0	No
Deep CNN [15]	Complex NN covert	10^{-2}	Sim	Yes	No	0	No
GAN power opt. [16]	Power theoretic	N/R	Sim	Yes	No	0	No
NRPCS [9]	Phase randomisation	N/R	Sim	No	No	2	No
SLTM [10]	Antenna sidelobe	N/R	Antenna	Partial	No	1	No
This work	8-domain EW evasion	0.0000	OTA 2.484 GHz	Yes	AES-128	8	$n=2000$

6. Further Insights and Analysis

370 6.1. Comparative Assessment

Table 9 positions the proposed system against representative covert and LPI communication approaches. Relative to the surveyed baselines, it is the only system that combines conditional GAN-based adaptive synthesis, zero BER under the reported impairment model, eight-domain EW resistance with
375 CNN detection below 60%, commercial-SDR OTA validation at 2.484 GHz, and quantified few-shot resistance.

6.2. Significance

The principal novelty of this work is the *simultaneous* satisfaction of all eight EW domain pass criteria within a single trained model, spanning

380 amplitude statistics, spectral flatness, cyclostationarity, higher-order moments,
 and time-frequency structure — each an independent detection surface that
 prior LPI systems address only partially. This is achieved through a nine-
 component composite loss that reinforces convergence toward AWGN statistics
 across all eight surfaces concurrently, combined with three anti-fingerprinting
 385 defences that prevent the generator from producing exploitable structural
 artefacts. Achieving all eight passes simultaneously under identical operating
 conditions (-5 dB SNR, 512 IQ samples/frame, 2.484 GHz) with commercial
 SDR hardware validation and quantified adversarial resistance (ROC AUC =
 0.583, few-shot detection $\leq 52.1\%$ at $n=2000$) represents, to the best of the
 390 authors' knowledge, a capability not demonstrated in the cited comparison
 literature under equivalent conditions. The few-shot result is also practically
 important: even with up to 2000 labelled intercepts, the adversarial detector
 remains below the conservative 70% operational reference threshold [4].

6.3. Limitations

- 395 1. **Fixed payload:** Each frame carries 336 bits; streaming voice or data
 requires multi-frame segmentation and reassembly.
2. **Reed–Solomon implementation:** A simplified parity encoder is used;
 a Berlekamp–Massey decoder would improve burst-error correction in
 production deployment.
- 400 3. **Single-carrier operation:** The waveform occupies a fixed 2 MHz at
 one carrier frequency; cognitive-radio frequency selection would improve
 resilience to frequency-selective EW.
4. **No frequency-selective multipath:** The training channel and OTA
 test were conducted under near line-of-sight (LoS) conditions; multipath

405 fading and Doppler effects have not been evaluated.

5. **Controlled OTA environment:** Testing was performed in a laboratory setting without co-channel Wi-Fi or Bluetooth interference.

6. **CNN detector training set:** The RadioML-style detector used 3200 balanced samples. Saturation was confirmed by repeating at
410 6400 samples ($\pm 0.8\%$ accuracy change), indicating the reported appreciation of 55.6% is not an underfitting artifact. More powerful classifiers or larger corpora may marginally raise the detection floor.

6.4. GAN Training Stability

GAN training is susceptible to mode collapse, whereby the generator
415 converges to a narrow output distribution irrespective of the latent input $\mathbf{z}$. Mode collapse would manifest here as near-identical output frames across different latent samples, which the diversity regularisation loss (Eq. 3) is specifically designed to prevent. Across all nine independent runs, the diversity loss remained active and positive throughout training, and D_1 remained within
420 49–52%, indicating that mode collapse did not occur. Training variability was low: RadioML detection rates spanned 51.5–57.9% and EW composite scores spanned 0.033–0.072 (all excellent) across nine runs, two messages, and two carrier frequencies, confirming reproducible stealth performance.

6.5. Future Work

425 Future extension possibilities include multi-frame streaming with CODEC2 or MELPe, longer-range field trials with co-channel interference, adaptive waveform reconfiguration driven by estimates of adversarial detector activity, multiple-input multiple-output (MIMO) spatial multiplexing for higher

throughput, and information-theoretic capacity analysis under the eight-
430 domain in-distinguishability constraint.

7. Conclusion

This paper presented a conditional GAN end-to-end system generating LPI communication waveforms that simultaneously evade eight independent EW detection domains while carrying AES-128-CTR-encrypted payloads. The nine-
435 component composite generator loss and three anti-fingerprinting defences produced zero BER from -10 to $+10$ dB SNR under a five-impairment RF channel model. An independently trained CNN detector achieved $55.6\% \pm 1.2\%$ balanced detection accuracy (95% CI: [54.74%, 56.46%]; ROC AUC = 0.583), confirming reduced detectability rather than undetectability under the tested
440 conditions, and the waveform remained below 52.1% detection across an eight-point few-shot experiment up to $n=2000$ adversarial training samples. A unified benchmarking comparison across six waveform families confirmed the highest composite stealth score of 0.888 among all evaluated communication waveform families, with only the AWGN reference baseline scoring higher
445 as non-communicating noise. OTA validation on a BladeRF Micro 2.0 at 2.484 GHz — transmitting from a Windows host, receiving on a separate Linux machine through real antennas — confirmed $BER=0.0000$ across 50 decoded windows with exact message recovery and decode confidence = 1.0. These results establish the practical feasibility of GAN-generated, multi-domain-EW-
450 evading waveforms for covert wireless communications in electromagnetically contested environments such as the 2.4 GHz ISM band.

CRediT Authorship Contribution Statement

Ambuj Sharma: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualisation. **Sumit**
455 **Gautam:** Supervision, Writing – review & editing, Resources. **Gaurav**
Sharma: Supervision, Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work
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470 NVIDIA T4 GPU platform.

Data Availability

The validated waveform files (`anshuman_lpi_waveform.npy`, `anshuman_lpi_complex.npy`), benchmark results (`ANSHUMAN_benchmark.csv`, `ANSHUMAN_complete_results.csv`), trained
475 model checkpoint (`anshuman_models.pt`), and the GNU Radio Companion flowgraph (`untitled.grc`) are available from the corresponding author upon reasonable request. The authors intend to deposit the codebase and datasets in a public repository (e.g. Zenodo or GitHub) upon acceptance.

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